

JBX-6300 series

OPERATIONS OF WRITING CONTROL PROGRAM

For the proper use of the instrument, be sure to read this instruction manual. Even after you read it, please keep the manual on hand so that you can consult it whenever necessary.

JBX-6300 series

OPERATIONS OF WRITING CONTROL PROGRAM

**Please be sure to read this instruction manual carefully,
and fully understand its contents prior to the operation
or maintenance for the proper use of the instrument.**

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MANUFACTURER

JEOL Ltd. 1-2, Musashino 3-chome, Akishima, Tokyo 196-8558 Japan
Telephone: 81-42-543-1111 Facsimile: 81-42-546-3353 URL: <http://www.jeol.co.jp>

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NOTATIONAL CONVENTIONS AND GLOSSARY

■ Examples for general notations

– CAUTION – :	Important precautions for use, which, if not followed, may result in damage to or problems with the device itself.
 :	Additional points to remember regarding the operation.
 :	A reference to another section, chapter or manual.
1, 2, 3 :	Numbers indicate a series of operations that achieve a task.
 :	A diamond indicates a single operation that achieves a task.
File :	The names of menus, commands, or parameters displayed on the screen are denoted with bold letters.
File–Exit :	Selecting a menu item from a pulldown menu is denoted by linking the menu and the item with a dash (–). For example, File–Exit means selecting Exit from the File menu.
 :	Keys on the keyboard are denoted by enclosing their names in a box.

■ Examples for mouse terminology

Mouse pointer:	A mark, displayed on the screen, which moves following the movement of the mouse. It is used to specify a menu item, command, parameter value, and other items. Its shape changes according to the situation.
Click:	To press and release the left mouse button.
Right-click:	To press and release the right mouse button.
Double-click:	To press and release the left mouse button twice quickly.
Drag:	To hold down the left mouse button while moving the mouse.

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1 GENERAL

This manual describes how to use the Writing Control program for the JEOL electron-beam lithography systems.

The main functions of the Writing Control program are shown below.

- Can be started from the EBX menu program.
 - Only one Writing Control program can be started from the EBX menu program. Exiting from the EBX menu program automatically terminates the Writing Control program, unless pattern writing is being performed.
- Lets you select a magazine file and perform pattern writing using the file.
- Provides the monitoring ability during pattern writing.
 - You can monitor the following.
 - Pattern-writing progress for each job
 - Path label, path number, and pattern name
 - Pattern-writing progress of chip arrays for each cassette
 - Field number
 - Data transfer for pattern writing
- Displays a summary of jobs specified in the magazine file before pattern writing.
- Displays the estimated pattern-writing time specified in the magazine file, if any.
- Lets you start the Stage Control program and move stage positions.
 - If the calibration function executed during pattern writing requires workpiece stages to be controlled for manual mark detection, you can start the Stage Control program from the Writing Control program and move stage positions. When you start pattern writing again, the Stage Control program automatically terminates. However, the Stage Control program started from the EBX menu program does not automatically terminate.
- Displays trend graphs of different kinds of calibration values during pattern writing.
- When you have started the Writing Control program on one display, you can start another Writing Control program on another display.
 - Another Writing Control program starts by duplicating the status of the Writing Control program that has been already started. For example, if the Writing Control program that has been already started is performing pattern writing when you start another Writing Control program, another Writing Control program starts by displaying the same progress as the program performing pattern writing, and then continues to display the progress.
 - When the Writing Control program that has been already started is performing pattern writing on one display, another Writing Control program cannot execute pattern writing.
- When pattern writing is completed, the Pattern writing completed window appears giving notice of completion.
 - The Pattern writing completed window also shows the number of cassettes in which any error occurred during pattern writing.
- Generates a pattern-writing log.

2 Starting the program and the EXPOSE window

2.1 Starting the Writing Control Program

To start the Writing Control program, follow the step below.

- ◆ In the EBX menu window, click on the **Expose** icon.

2.2 Expose Window

When you start the Writing Control program, the Expose window appears.

Immediately after you start the program, no information appears in the progress area because pattern writing has not started yet. When you start to perform pattern writing, the progress area starts to display pattern-writing progress.

If pattern writing is proceeding on one display when you start another Writing Control program, another Writing Control program starts to display pattern-writing progress in the progress area at the time.

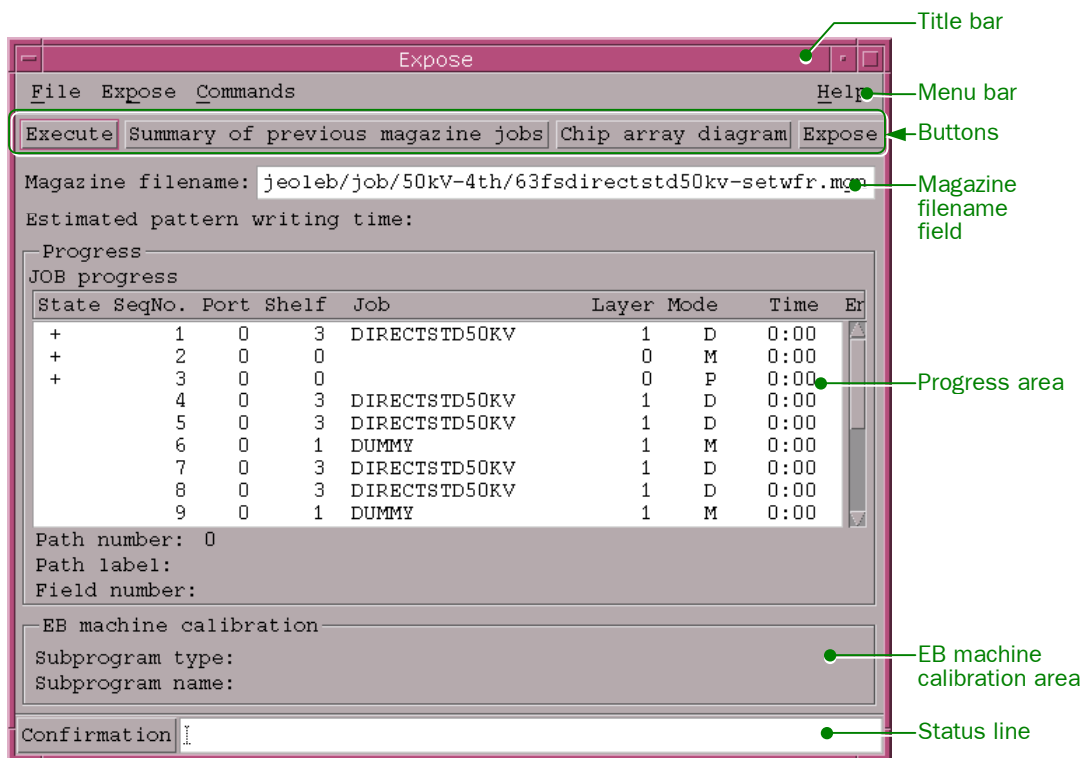


Fig. 1 Expose Window

2.2.1 Title bar

The title bar would slightly differ depending on the window system under which the program runs. Follow the directions for use of your window system.

2.2.2 Menu bar

The menu bar includes the following four menus.

- **File**
- **Expose**
- **Commands**
- **Help**

The commands that the menus contain are shown below.

● **File**

The **File** menu contains the following commands.

Magazine File: Lets you select the magazine file to be used for pattern writing.

Exit: Terminates the program.

● **Expose**

The **Expose** menu contains the following commands.

Execute: Starts pattern writing using the selected magazine file.

Stop: Stops pattern writing.

● **Commands**

The **Commands** menu contains the following command.

Trend display: Lets you select to display a trend graph of one of the following subprograms.

CURRNT

DRIFT

PDEFBE

INITAE

INITBE

● **Help**

The **Help** menu contains the following command.

Product Information:

Displays the product version number and the date and time at which the program was created (compiled).

2.2.3 Buttons

The **Expose** window provides the following buttons.

Execute: Starts pattern writing using the selected magazine file.

Chip array diagram:

Displays the chip array diagram for each cassette for which pattern writing is proceeding.

Job summary of the previous magazine

Currently unused.

Expose log display

Displays a pattern-writing log.

2.2.4 Magazine filename field

The Magazine filename field displays the name of the selected magazine file and the estimated pattern-writing time specified in the magazine file.

2.2.5 Progress area

The progress area displays pattern-writing progress, as shown below.

● Job progress list

While pattern writing is proceeding, the progress area displays a list of jobs specified in the selected magazine file with progress for each job.

When you select a job from the list of jobs, the Job summary window appears.

Different lines of the list correspond to different jobs. A line consists of the following information.

- State
Status of the job: To be done (*) | In progress (>>) | Done (+) | Unsuccessful cassette pattern writing (-)
- SeqNo.
Job number starting with 1
- Port
Currently unused. It always displays 0.
- Shelf
Shelf number of carrier
- Job
Jobdeck name
- Layer
Layer number
- Mode
Pattern-writing mode: M | P | D
M: Regular writing on mask
D: Direct writing
- Time (Currently unused.)
Estimated pattern-writing time
- Ending time
When pattern writing starts or is proceeding, the time calculated by adding the estimated pattern-writing time to the pattern-writing starting time appears. As pattern writing ends, the time at which pattern writing actually ended appears.

● Path number

The number of the path for which pattern writing is proceeding appears.

● Path label

The label of the path for which pattern writing is proceeding appears.

● Field number

While pattern writing is proceeding, the field number is updated accordingly.

2.2.6 EB machine calibration area

The **EB machine calibration** area displays the following information about calibration being executed during pattern writing.

- **Subprogram type**

The type of the subprogram being executed, including Initial and Cyclic, appears.

- **Subprogram name**

The name of the subprogram being executed appears.

2.2.7 Status line

When errors occur, an error message appears in the Status line.

3 Using pattern-writing function

3.1 Selecting Magazine File

To select the magazine file with which to perform pattern writing, follow the steps below.

1. In the Expose window, choose **File – Magazine File**.

The Magazine file selection window will appear.

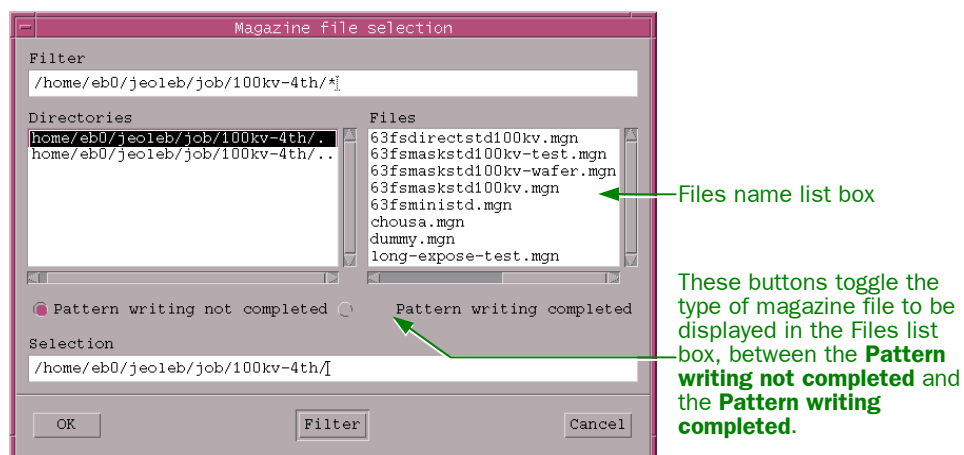


Fig. 2 Magazine file selection window

2. In the Files list box, double-click on the file name, or click on the file name and then the **OK** button.

The selected magazine file name will appear in the Magazine filename field in the Expose window.

3.2 Performing Pattern Writing

To perform pattern writing, follow the steps below.

1. In the **Expose** window, click on the **Execute** button or choose **Expose – Execute**.

The Pattern writing execution check window will appear showing the selected magazine file name and the jobs that the file specifies.

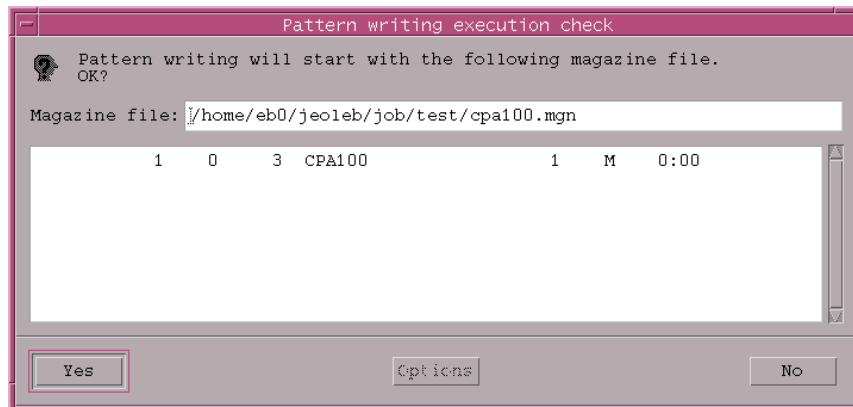


Fig. 3 Pattern writing execution check window

2. Make sure that the displayed magazine file name is right; then click on the **Yes** button.

Pattern writing will start.

3.3 Pattern-Data Transfer

When pattern-data transfer for pattern writing starts, the HSC transfer progress window appears. When pattern-data transfer is completed, the HSC transfer progress window automatically closes.

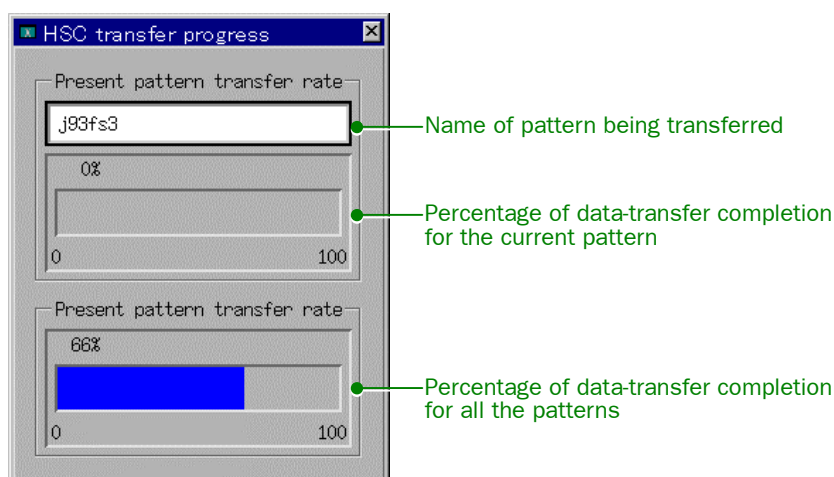


Fig. 4 HSC transfer progress window

In some cases, the transfer progress may appear by transfer categories, that is, by the magazine, by the cassette, and by the path. If all the patterns are for field-shift pattern writing, the HSC transfer progress window does not appear.

3.4 Displaying Chip Array Diagram

To display the chip array diagram, follow the step below.

- ◆ During pattern writing, click on the **Chip array diagram** button in the Expose window.

The Chip array diagram window will appear showing the pattern name and pattern-writing progress of the chips.

In case of wafer pattern writing, a window resembling Fig. 5 appears. In case of mask pattern writing, a window resembling Fig. 6 appears.

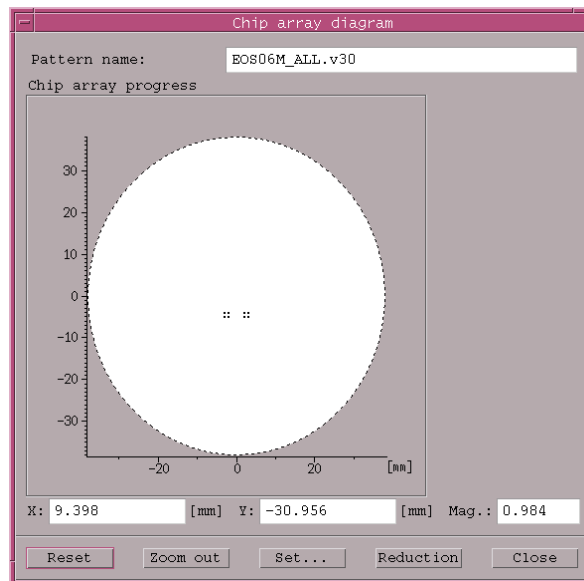


Fig. 5 Chip array diagram: Pattern writing on wafer

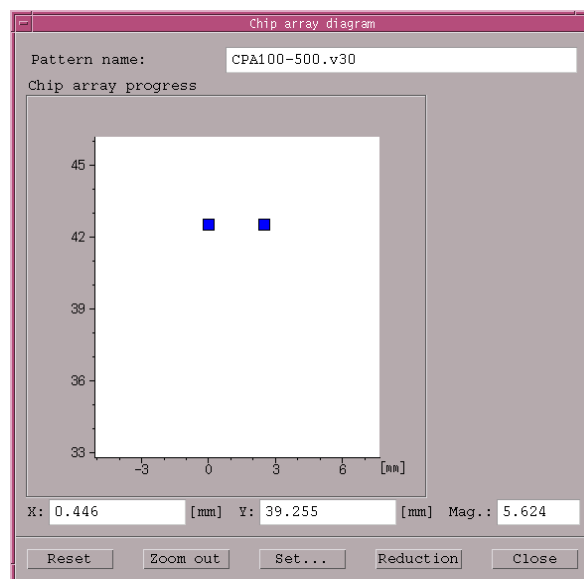


Fig. 6 Chip array diagram: Pattern writing on mask

You can identify pattern-writing progress for a chip by the following colors.

- To be done: Outline only

- In progress: Yellow
- Done Normal end: Blue
 Abnormal end: Red

1. Adjust the display appropriately by using the **Set ... button or the **Reduction** button.**

Set... button You can specify the display area.

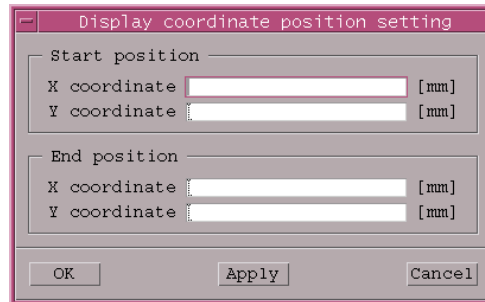


Fig. 7 Display coordinate position setting window

An error message appears when you enter a setting that is outside the display area (Fig. 8).

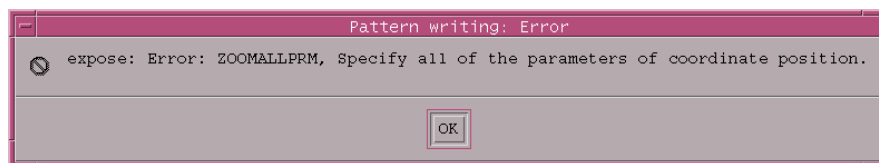


Fig. 8 pattern writing: Error window

Reduction button Clicking this button reduces the display (Fig. 9).

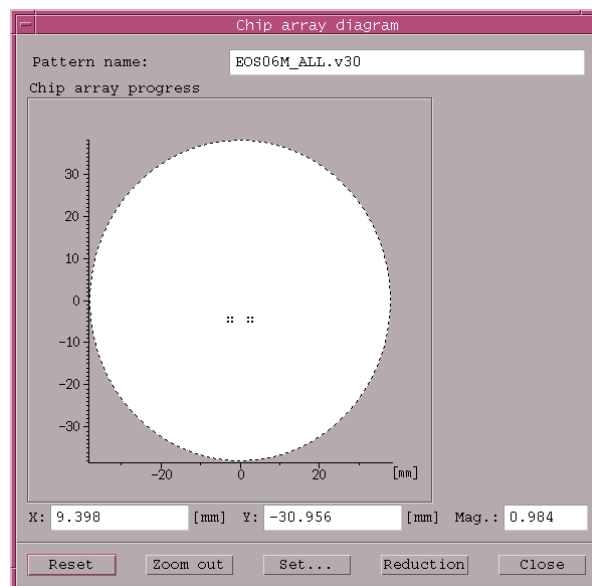


Fig. 9 Chip array diagram

Zoom out button Clicking this button shows the previous display (Fig. 1).
Reset button Clicking this button shows the standard display (Fig. 5, Fig. 6).

3.5 Displaying Trend Graph during Pattern Writing

When you choose **Commands – Trend display** while pattern writing is proceeding, and when you choose a subprogram name from the submenu, the trend graph of the chosen subprogram appears. Table 1 shows the trend graphs that you can choose and their descriptions.

Table 1 Types of trend graphs

Subprogram	Horizontal scale	Vertical scale	Type of graph	Graph name
CURRNT	Minute	pA	current setting	CURRNT
DRIFT	Minute	nm	BE mark position X BE mark position Y X drift Y drift	DRIFT-X DRIFT-Y DRIFT-X/CASSETTE DRIFT-Y/CASSETTE
PDEFBE	Minute	point, number of times	Gain correction value X, Y Rotation correction value X, Y Number of convergence	PDEFBE-GAIN PDEFBE-ROTATION PDEFBE adjustment times
INITAE	Minute	nm, number of times	Beam position X, Y Beam size X, Y Number of convergence	Beam Position Beam Size Convergent Count
INITBE	Minute	nm, S/N ratio	Mark position X, Y Mark size X, Y S/N ratio	Mark Position Mark Width S/N Rate

3.5.1 Trend (CURRNT) window

To display the trends of CURRNT values, choose **Commands – Trend display – CURRNT** while pattern writing is proceeding. Then the Trend (CURRNT) window appears showing the trends of CURRNT values.

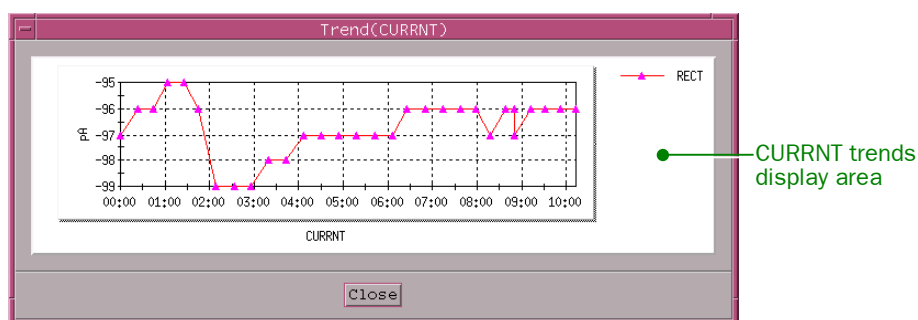


Fig. 10 Trend (CURRNT) window

CURRNT trends display area

The vertical axis shows pA. The horizontal axis shows time.

The plot shows raw data.

The Trend (CURRNT) window displays data from the pattern-writing start time to the current time. The horizontal scale is optimized depending on the current time, beginning with 0 for time at which the first measurement for each subprogram is performed.

This graph plots data each time the current is measured.

3.5.2 Trend (DRIFT) window

To display the trends of DRIFT values, choose **Commands – Trend display – DRIFT** while pattern writing is proceeding. Then the Trend (DRIFT) window appears showing the trends of DRIFT values.

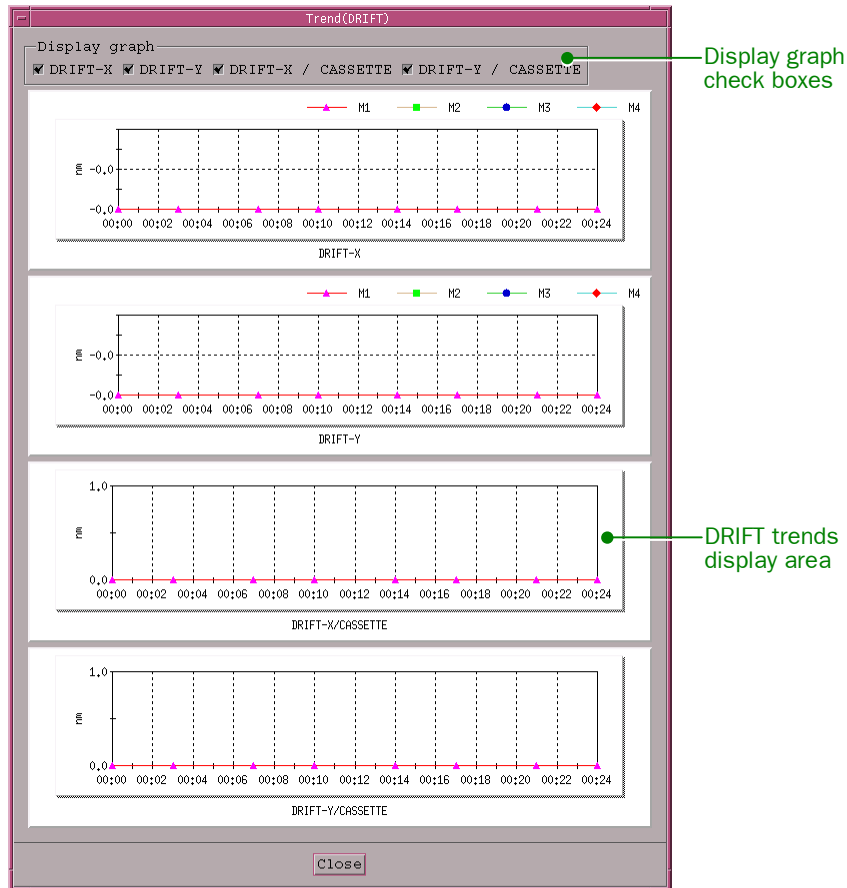


Fig. 11 Trend (DRIFT) window

Display graph check boxes

When a check box is checked, the graph for the check box appears. By default, all the boxes are checked.

The DRIFT-X and DRIFT-Y graphs plot raw data. The DRIFT-X/CASSETTE and DRIFT-Y/CASSETTE graphs show drifts for every cassette, using 0 for the initial value.

DRIFT trends display area

The vertical scale is in nm. The horizontal scale is a time scale. The horizontal scale is optimized depending on the current time, beginning with 0 for the pattern-writing start time.

3.5.3 Trend (PDEFBE) window

To display the trends of PDEFBE values, choose **Commands – Trend display – PDEFBE** while pattern writing is proceeding. Then the Trend (PDEFBE) window appears showing the trends of PDEFBE values.

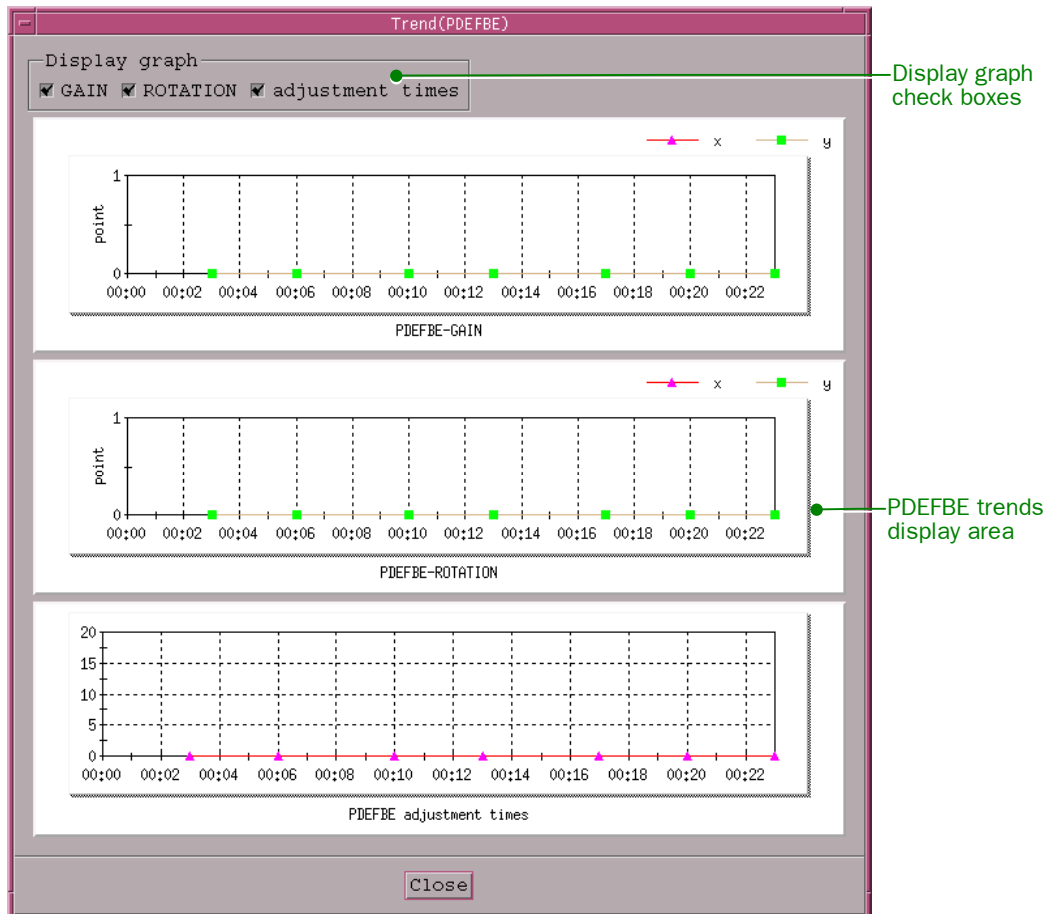


Fig. 12 Trend (PDEFBE) window

Display graph check boxes

When a check box is checked, the graph for the check box appears.

By default, all the boxes are checked.

The Trend (PDEFBE) window can provide three types of graphs; GAIN, ROTATION, and adjustment times.

PDEFBE trends display area

The vertical scales of the GAIN and ROTATION graphs are in point (convergence value). The vertical scale of the adjustment times graph is in number of times.

The horizontal scale is a time scale. The horizontal scale is optimized depending on the current time, beginning with 0 for the pattern-writing start time.

3.5.4 Trend (INITAE) window

To display the trends of INITAE values, choose **Commands – Trend display – INITAE** while pattern writing is proceeding. Then the Trend (INITAE) window appears showing the trends of INITAE values.

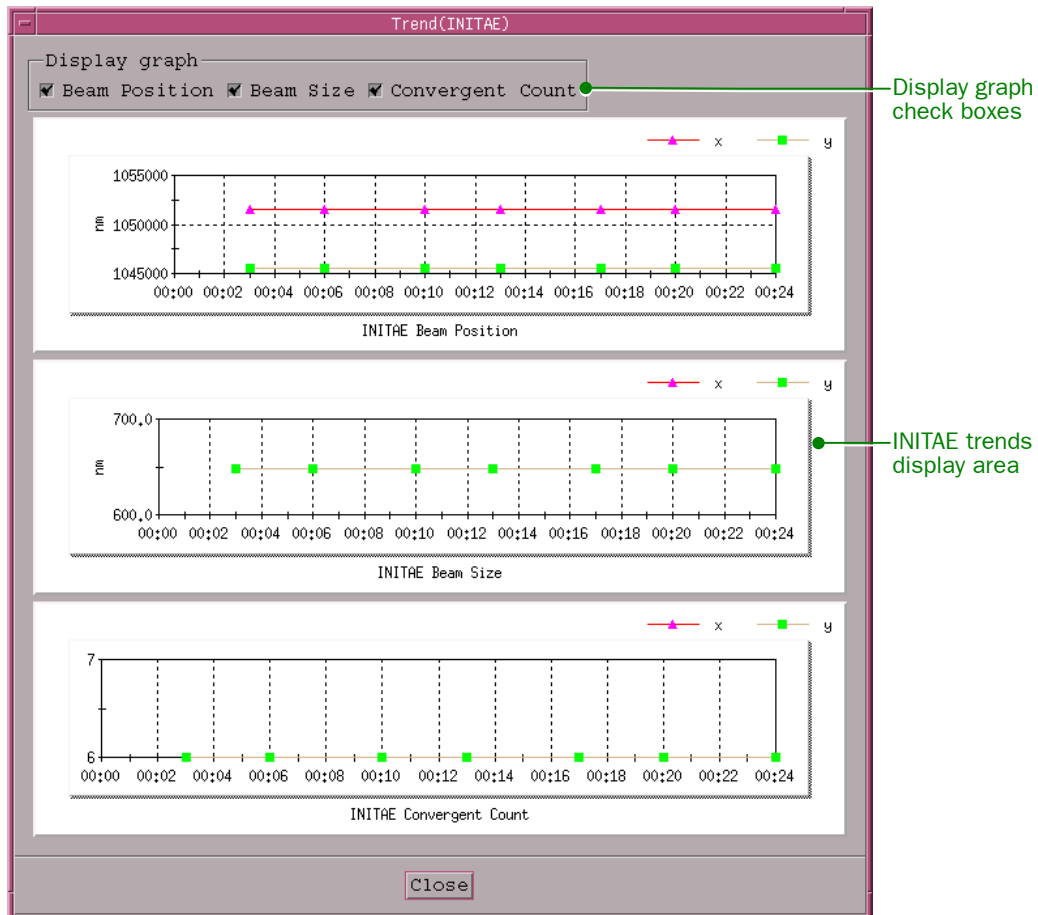


Fig. 13 Trend (INITAE) window

Specified region of the graph

When a check box is checked, the graph for the check box appears. By default, all the boxes are checked.

The Trend (INITAE) window can provide four types of graphs; Beam Position, Beam Size, Edge Sharpness, and Convergent Count.

INITAE trends display area

The vertical scales of the Beam Position, Beam Size, and Edge Sharpness graphs are in nm. The vertical scale of the Convergent Count graph is in number of times.

The horizontal scale is a time scale. The horizontal scale is optimized depending on the current time, beginning with 0 for the pattern-writing start time.

3.5.5 Trend (INITBE) window

To display the trends of INITBE values, choose **Commands – Trend display – INITBE** while pattern writing is proceeding. Then the Trend (INITBE) window appears showing the trends of INITBE values.

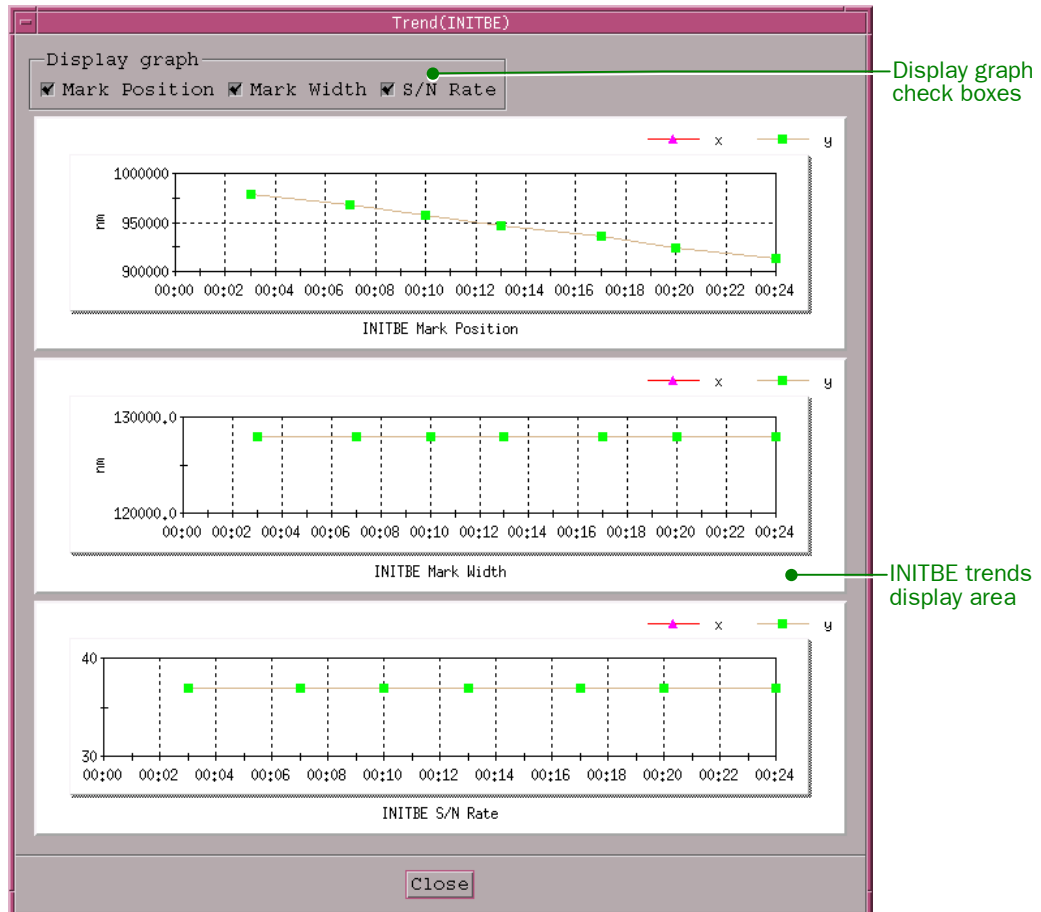


Fig. 14 Trend (INITBE) window

Display graph check boxes

When a check box is checked, the graph for the check box appears. By default, all the boxes are checked.

The Trend (INITBE) window can provides three types of graphs; Mark Position, Mark Width, and S/N Rate.

INITBE trend display area

The vertical scales of the Mark Position and Mark Width graphs are in nm. The horizontal scale is a time scale. The horizontal scale is optimized depending on the current time, beginning with 0 for the pattern-writing start time.

3.6 Mark Detection

3.6.1 Mark search detection window

If SETWFR or PSMASK executed in the semi-automatic mode fails in mark detection during pattern writing, the Mark search detection window automatically appears.

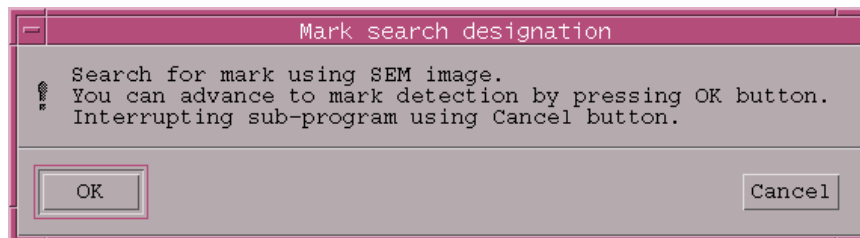


Fig. 15 Mark search detection window

Button	Function
OK	Starts the Stage Control program and opens the Mark position setting check window. The Image Display program will change the SEM image display to the waveform display.
Cancel	Does not perform mark detection and stops pattern writing. The Auto Loader program and the Image Display program started by performing pattern writing will automatically terminate.

3.6.2 Mark detection execution check window

If you click on the **OK** button in the Mark search detection window, the Mark detection execution check window appears.

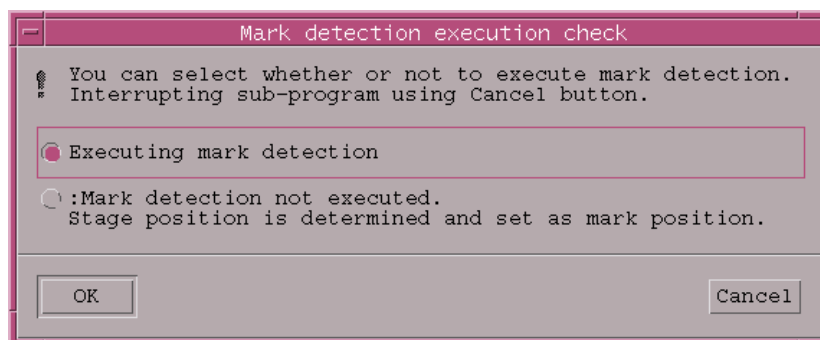


Fig. 16 Mark detection execution check window

Button	Function
OK	Continues pattern writing. The Stage Control program started by clicking on the Yes button in the Mark search detection window will terminate. The Image Display program will change the SEM image display to the waveform display.
Cancel	Stops pattern writing. The Stage Control program started by clicking on the Yes button in the Mark search detection window will terminate.

3.7 Stopping Pattern Writing

To stop pattern writing, follow the steps below.

1. Choose **Expose – Stop** in the **Expose** window.

The Pattern writing stop window will appear.

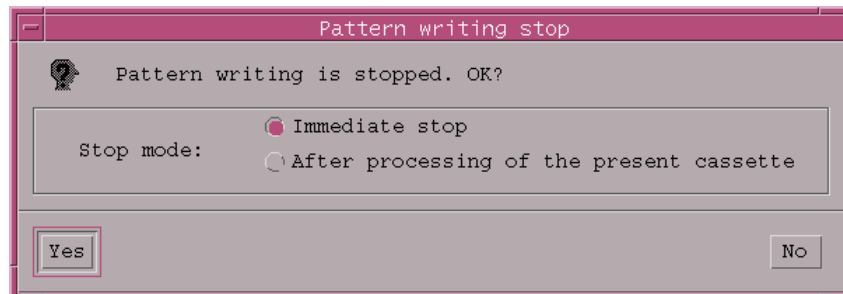


Fig. 17 Pattern writing stop window

2. Select either the **Immediate stop** or the **After processing of the present cassette** stop mode.
3. Click on the **Yes** button.

Pattern writing will stop in the selected stop mode.

3.8 Pattern Writing Completed Window

When pattern writing is completed, the Pattern writing completed window automatically appears giving notice of completion. This window also shows the number of cassettes in which any error occurred.

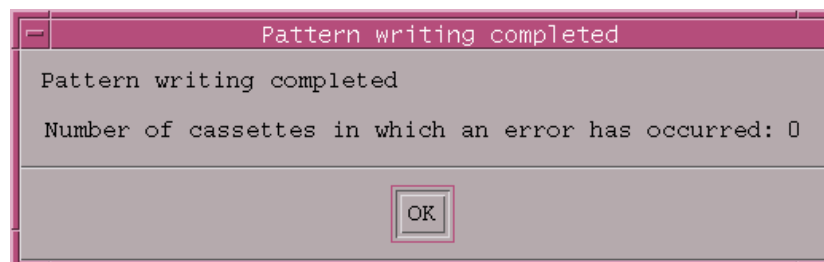


Fig. 18 Pattern writing completed window

4 Pattern-writing log

4.1 Displaying Pattern-Writing Log

When you perform pattern writing, a pattern-writing log is automatically generated. To see a pattern-writing log, follow the steps below.

1. In the Expose window, click on the **Expose log display** button.
The Expose Log window will appear.

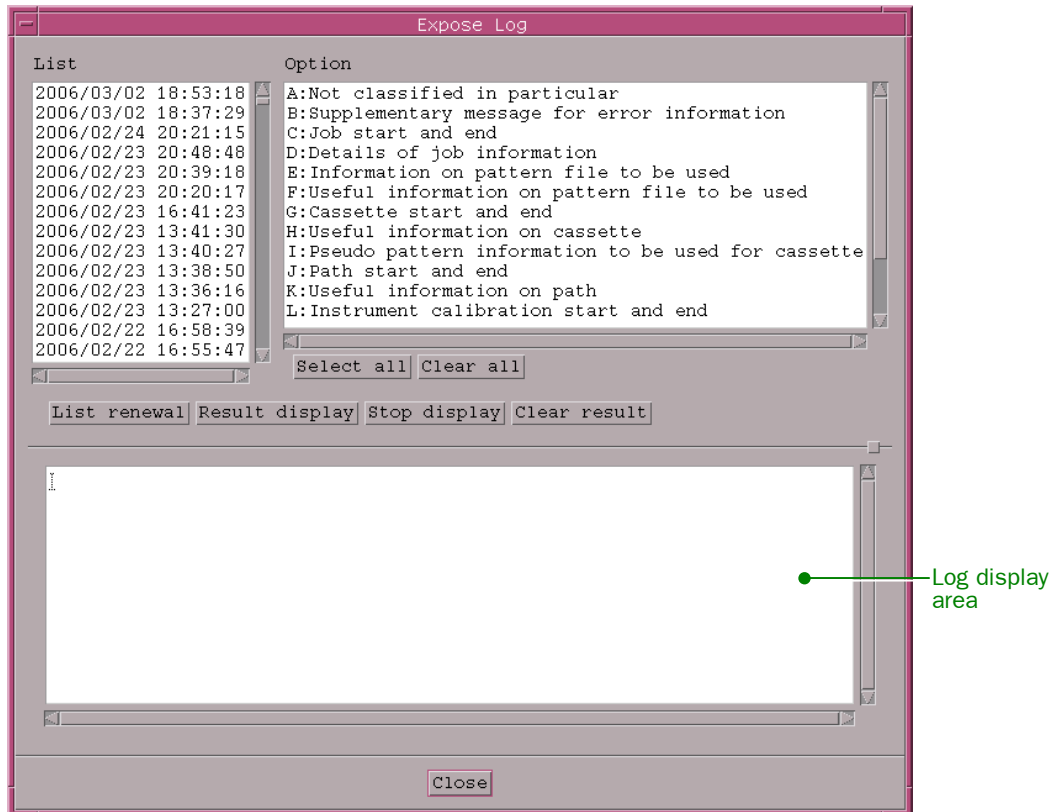


Fig. 19 Expose Log window

Item	Function
List box	Lists generated pattern-writing logs. From the list, select a log you want to display. You can select multiple logs.
Option box	Lists options that specify log items to be displayed. From the list, select a option you want to use. You can select multiple options.
Select all button	Selects all the options in the Option list box.
Clear all button	Deselects all the options in the Option list box.
List renewal button	Searches the directory containing pattern-writing logs for a new log and updates the List box.
Result display button	Displays the selected log in the log display area using the selected options. This button is disabled when there is a log displayed in the log display area.
Stop display button	Stops displaying a log. This button is disabled when there is no log displayed in the log display area.
Clear result button	Clears the log display area.
Log display area	When you click on the Result display button, this area displays the selected log.
Close button	Closes this window.

2. From the List box, select one or more logs you want to display.
3. From the Option list box, select one or more options with which to specify log items to be displayed.
4. Click on the **Result display** button.
The selected log will appear in the log display area.

5 Help function

5.1 Displaying Product Information

To see the product information, follow the step below.

- ◆ In the Expose window, choose **Help – Product Information**.

The Product Information window appears showing the product version number and the date and time at which the program was created (compiled).

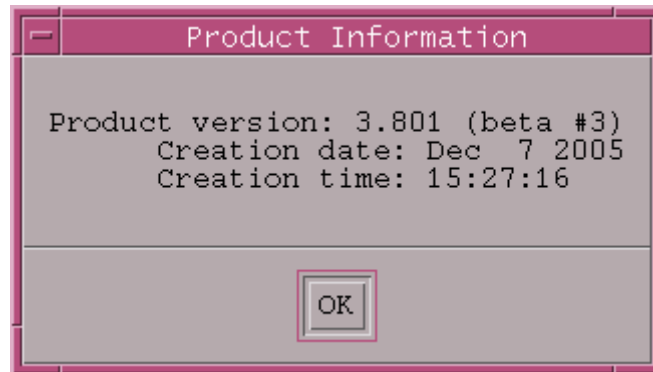


Fig. 20 Product Information window